

Answer the marked questions

Individual works with Simulator

1.	Design a 4 bit full adder with necessary basic circuit element.	4
2.	Design a BCD to Excess -3 code converter.	4
3.	Design a 4 bit comparator circuit.	4
4.	Design a 4 bit even parity generator circuit.	
✓ 5.	Design a 4 -bit Decimal adder.	
6.	Design a Hexadecimal to Binary encoder with necessary basic circuit element.	
7.	Design MUX and DMUX in one board to test the full MUX-DMUX operation.	
8.	Design a D flip-flop.	

Group Works

9	A minority gate is a digital circuit whose output is equal to 1 if the minorities of inputs are 1's. The output is 0 otherwise. By means of a truth table, find the Boolean function implemented by a 3-input minority gate. Simplify the function.
10	Simplify $F = \Sigma (0, 1, 4, 5, 7, 9, 13)$ into sum-of-products form implement both of its form with necessary logic gates.
11	Simplify $F = \Sigma (0, 1, 4, 5, 7, 9, 13)$ into product-of-sums form and implement both of its form with necessary logic gates.
12	Let $F = A'C + A'B + AB'C + BC$ is a Boolean expression a) Express it in sum of minterms b) Find the minimal sum of products expression and implement both of its form with necessary logic gates.
13	Viva-Voce



Patuakhali Science and Technology University (PSTU)
Department of Computer Science and Information Technology (CIT)

Final Examination (LAB), July-December-2018

Program: B. Sc. Engg. (CSE)

Session: 2016-17

Course Code : CCE-221

Course Title : Digital Logic Design Lab

Full Time: 3 hours

Full Marks: 70

Answer the two marked questions from set A and B; Set C is compulsory.

Set A

- | | | |
|----|---|----|
| 1. | A minority gate is a digital circuit whose output is equal to 1 if the minorities of the inputs are 1's. The output is 0 otherwise. By means of a truth table, find the Boolean function implemented by a 3-input minority gate. Simplify the function. | 20 |
| 2. | Given the function $F(W, X, Y, Z) = \sum m(1, 5, 7, 8, 9, 15) + d(4, 14)$ find the minimum Sum-of-Products using Q & M method. Show your work including the charts. | 20 |
| 3. | Simplify $F = (A' + B + C) (B' + C + D) (A' + B' + C' + D) (A + B' + C)$ and implement it with necessary logic gates. | 20 |
| 4. | Simplify $F(A, B, C, D) = \Pi(0, 2, 3, 5, 7, 8, 9, 10, 11, 13, 15)$ and implement it with necessary logic gates. | 20 |
| 5. | Simplify $F = \sum (0, 1, 4, 5, 7, 9, 13)$ into (a) sum-of-products form, and (b) product-of-sums form and implement both of its form with necessary logic gates. | 20 |

Set B

- | | | |
|-----|---|----|
| 6. | Design an AND-to-OR gate combinational network for the Boolean algebra function:
$F = W'X'Y'Z' + WX'Y'Z + WX'Y'Z' + W'XY'Z'$ Use as few gates as possible. | 30 |
| 7. | Design an 8 bit BCD adder. | 30 |
| 8. | Design an 8 bit parity checker. | 30 |
| 9. | Design an 8 bit full adder with sequential logic procedure. | 30 |
| 10. | Design an 8 bit registrar with parallel load. | 30 |

Set C

- | | | |
|-----|-----------|----|
| 11. | Viva-Voce | 20 |
|-----|-----------|----|

DLD Question.pdf

simplify the Boolean function

$$F(w, x, y, z) = \sum(1, 3, 7, 11, 15)$$

which has the don't-care conditions

$$d(w, x, y, z) = \sum(0, 2, 5)$$

and show "An expression with the minimum number of literals is not necessarily unique".

What are the purposes of arithmetic circuits? Design and describe about a 4-bit full adder with a look-ahead carry generator.

Design a logic circuit to determine a prime number ranging from 0 to 15.

$$\begin{array}{r} 1001 \\ 0011 \\ \hline 1100 \end{array}$$

Patuakhali Science & Technology University (PSTU)
Department of Computer Science and Information Technology (CSIT)

Final Examination: July-December 2020

Course Code: CCE 222 | Course Title: Digital Logic Design Sessional

Session: 2018-19, Program: B.Sc. Engg.(CSE), Semester: 4th

Marks - 70

[Answer the marked questions]

Simulation - 10/15

Section A (Use Simulator to Implement)

1. Design a combinational circuit which accepts a two-bit number and generates an output binary number equal to the cube of the input number. 20
2. Implement following Boolean function using exclusive or and AND gates 20
 $F = ABCD + A'B'C'D' + ABC'D' + A'B'CD$
3. With 2×1 mux implement XOR gate and AND gate. 20
4. Design a D Flip-flop. 20
5. Design a T Flip-flop. 20
6. Design a circuit that compares two 4-bit numbers, A and B, to check if they are equal. The circuit has one output x, so that $x=1$ if $A=B$, and $x=0$ if $A \neq B$. 20

Section B (Use Circuit board to Implement by group work)

7. Design an adder that can add two decimal numbers as following. 30

$$\begin{array}{r} ABCD \\ + PQRS \\ \hline WXYZ \end{array}$$

8. Viva-Voce 20

Patuakhali Science & Technology University (PSTU)
Department of Computer Science and Information Technology (CSIT)

Mid Term Examination: July-December 2022

Course Code: CCE 222 | Course Title: Digital Logic Design Sessional

Session: 2020-21, Program: B.Sc. Engg.(CSE), Semester: 4th

Marks – 15

[Answer all the questions]

1.	Design a circuit that produce output that square of the given input. Use 3 bit input to implement it with NAND gate.	15
✓ 2.	Simplify $F = A'B'C' + AB'D + A'B'CD'$ and implement it with necessary NOR gates.	15
3.	Simplify $F = w'xz + w'yz + x'yz' + wxyz$ $d = wxyz$ and implement it with necessary NAND gates.	15
4.	Simplify $F(A, B, C, D) = \Pi(0, 2, 3, 5, 7, 8, 9, 10, 11, 13, 15)$ and implement it with necessary logic gates.	15
5.	Simplify $F = \Sigma(0, 1, 4, 5, 7, 9, 13)$ into (a) sum-of-products form, and (b) product-of-sums form and implement both of its form with necessary logic gates.	15
6.	Show that $A \oplus B \oplus C \oplus D = \Sigma(0, 3, 5, 6, 9, 10, 12, 15)$. and implement it.	15
7	Design a 4 bit full adder .	15

Patuakhali Science & Technology University (PSTU)
Department of Computer Science and Information Technology (CSIT)

Final Examination: July-December 2020

Course Code: CCE 222 | Course Title: Digital Logic Design Sessional

Session: 2018-19, Program: B.Sc. Engg.(CSE), Semester: 4th

Marks – 70

[Answer the marked questions]

Simulation - 10:15

Section A (Use Simulator to Implement)

1. Design a combinational circuit which accepts a two-bit number and generates an output binary number equal to the cube of the input number. 20
2. Implement following Boolean function using exclusive or and AND gates 20
 $F = ABCD + A'B'C'D' + ABC'D' + A'B'CD$
3. With 2×1 mux implement XOR gate and AND gate. 20
4. Design a D Flip-flop. 20
5. Design a T Flip-flop. 20
6. Design a circuit that compares two 4-bit numbers, A and B, to check if they are equal. The circuit has one output x, so that $x=1$ if $A=B$, and $x=0$ if $A \neq B$. 20

Section B (Use Circuit board to Implement by group work)

7. Design an adder that can add two decimal numbers as following. 30

$$\begin{array}{r} A B C D \\ + P Q R S \\ \hline W X Y Z \end{array}$$

8. Viva-Voce 20

2025/01/14 11:35

Patuakhali Science & Technology University (PSTU)
Department of Computer Science and Information Technology (CSIT)

Final Examination: July-December 2021

Course Code: CCE 222 | Course Title: Digital Logic Design Sessional

Session: 2019-20, Program: B.Sc. Engg.(CSE), Semester: 4th

Marks – 70

Section A [Use Circuit board to implement any questions by group work]

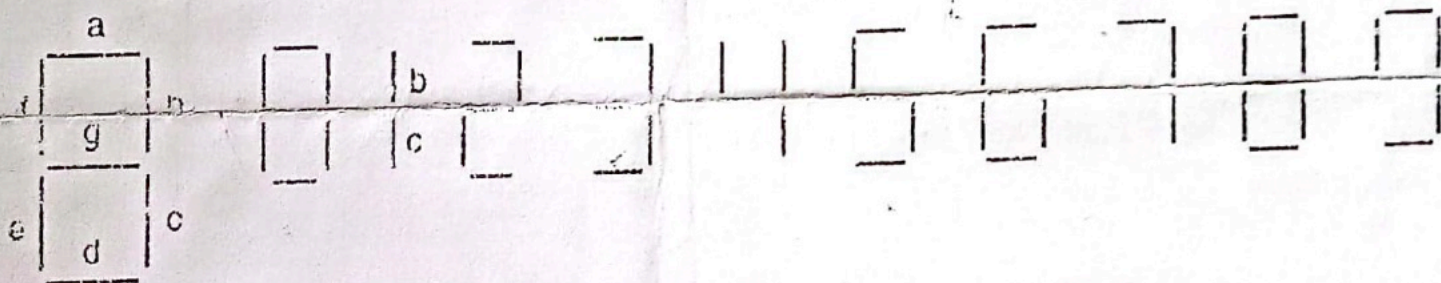
1. Implement following Boolean function using NAND or NOR gates

$$F = ABCD + A'B'C'D' + ABC'D' + A'B'CD$$

2. Design a circuit that compares two 4-bit numbers, A and B, to check if they are equal. The circuit has one output x, so that $x = 1$ if $A = B$, and $x = 0$ if $A \neq B$.

Section B (Use Circuit board to implement by group work)

3. Design a BCD-to-seven-segment decoder is a combinational circuit that accepts a decimal digit in BCD and generates the appropriate outputs for selection of segments in a display indicator used for displaying the decimal digit. The seven outputs of the decoder (a, b, c, d, e, f, g) select (a) Segment designation



(b) Numerical designation for display

the corresponding segments in the display as shown in Fig. (a). The numeric designation chosen to represent the decimal digit is shown in Fig. (b). Design the BCD-to-seven-segment decoder circuit.

4. Lab Report

8. Viva-Voce

2025/01/14 11:35

10
15

Patuakhali Science and Technology University
Department of Computer Science and Information Technology

4th Semester (Level-2, Semester-I), Midterm Examination of B.Sc. Engg.(CSE), July-December/2021, Session: 2019-20

Course Code: CCE-221 Course Title: Digital Logic Design

Full Marks: 15 Duration: 45 minutes

[Figures in the right margin indicate full marks]

Answer all the following questions.

1. Simplify the Boolean function 5
$$F(w, x, y, z) = \sum(1, 3, 7, 11, 15)$$

which has the don't-care conditions
$$d(w, x, y, z) = \sum(0, 2, 5)$$

and show "An expression with the minimum number of literals is not necessarily unique".
2. What are the purposes of arithmetic circuits? Design and describe about a 4-bit full adder with a look-ahead carry generator. 5
- Design a logic circuit to determine a prime number ranging from 0 to 15. 5

Patuakhali Science & Technology University (PSTU)
Department of Computer Science and Information Technology (CSIT)

Mid Term Examination: July-December 2021
Course Code: CCE 222 | Course Title: Digital Logic Design Sessional
Session: 2019-20, Program: B.Sc. Engg.(CSE), Semester: 4th

Marks - 15

[Answer all the questions]

1. A minority gate is a digital circuit whose output is equal to 1 if the minorities of the inputs are 1's. The output is 0 otherwise. By means of a truth table, find the Boolean function implemented by a 3-input minority gate. Simplify the function. 15
2. Simplify $F(w, x, y, z) = \sum (0, 1, 2, 4, 7, 8, 9, 12, 13, 15)$ and implement it with necessary logic gates. 15
3. Simplify $F = (A' + B + C) (B' + C + D) (A' + B' + C' + D) (A + B' + C)$ and implement it with necessary logic gates. 15
4. Simplify $F(A, B, C, D) = \Pi(0, 2, 3, 5, 7, 8, 9, 10, 11, 13, 15)$ and implement it with necessary logic gates. 15
5. Simplify $F = \sum (0, 1, 4, 5, 7, 9, 13)$ into (a) sum-of-products form, and (b) product-of-sums form and implement both of its form with necessary logic gates. 15
6. Design a Full-Subtractor. 15
7. Design a 4 bit full adder with look ahead carry generator. 15
9. Design a BCD adder 15

2025/01/14 11:36

Patuakhali Science & Technology University (PSTU)
Department of Computer Science and Information Technology (CSIT)

Mid Term Examination: July-December 2022

Course Code: CCE 222 | Course Title: Digital Logic Design Sessional

Session: 2020-21, Program: B.Sc. Engg.(CSE), Semester: 4th

Marks – 15

[Answer all the questions]

1.	Design a circuit that produce output that square of the given input. Use 3 bit input to implement it with NAND gate.	15
✓ 2.	Simplify $F = A'B'C' + AB'D + A'B'CD'$ and implement it with necessary NOR gates.	15
3.	Simplify $F = w'xz + w'yz + x'yz' + wx'y'z$ $d = w y z$ and implement it with necessary NAND gates.	15
4.	Simplify $F(A, B, C, D) = \Pi(0, 2, 3, 5, 7, 8, 9, 10, 11, 13, 15)$ and implement it with necessary logic gates.	15
5.	Simplify $F = \Sigma(0, 1, 4, 5, 7, 9, 13)$ into (a) sum-of-products form, and (b) product-of-sums form and implement both of its form with necessary logic gates.	15
6.	Show that $A \oplus B \oplus C \oplus D = \Sigma(0, 3, 5, 6, 9, 10, 12, 15)$. and implement it.	15
7	Design a 4 bit full adder .	15

Patuakhali Science & Technology University (PSTU)
Department of Computer Science and Information Technology (CSIT)

Mid Term Examination: July-December 2021

Course Code: CCE 222 | Course Title: Digital Logic Design Sessional

Session: 2019-20, Program: B.Sc. Engg.(CSE), Semester: 4th

Marks - 15

[Answer all the questions]

1. A minority gate is a digital circuit whose output is equal to 1 if the minorities of the inputs are 1's. The output is 0 otherwise. By means of a truth table, find the Boolean function implemented by a 3-input minority gate. Simplify the function. 15
2. Simplify $F(w, x, y, z) = \sum (0, 1, 2, 4, 7, 8, 9, 12, 13, 15)$ and implement it with necessary logic gates. 15
3. Simplify $F = (A' + B + C) (B' + C + D) (A' + B' + C' + D) (A + B' + C)$ and implement it with necessary logic gates. 15
4. Simplify $F(A, B, C, D) = \Pi(0, 2, 3, 5, 7, 8, 9, 10, 11, 13, 15)$ and implement it with necessary logic gates. 15
5. Simplify $F = \sum (0, 1, 4, 5, 7, 9, 13)$ into (a) sum-of-products form, and (b) product-of-sums form and implement both of its form with necessary logic gates. 15
6. Design a Full-Subtractor. 15
7. Design a 4 bit full adder with look ahead carry generator. 15
9. Design a BCD adder. 15

1702064

Patuakhali Science and Technology University (PSTU)

Department of Computer Science and Communication Engineering (CSE)

Final Examination (LAB), July-December: 2019

Program: B. Sc. Engg.(CSE)

Session: 2017-18

Course Code : CCE-222

Course Title : Digital Logic Design Lab

Full Time: 3 hours

Full Marks: 70

Answer the marked questions

Individual works with Simulator

1.	Design a 4 bit full adder with necessary basic circuit element.	40
2.	Design a BCD to Excess -3 code converter.	40
3.	Design a 4 bit comparator circuit.	40
4.	Design a 4 bit even parity generator circuit.	40
5.	Design a 4 -bit Decimal adder.	40
6.	Design a Hexadecimal to Binary encoder with necessary basic circuit element.	40
7.	Design MUX and DMUX in one board to test the full MUX-DMUX operation.	40
8.	Design a D flip-flop.	40

Group Works

9.	A minority gate is a digital circuit whose output is equal to 1 if the minorities of the inputs are 1's. The output is 0 otherwise. By means of a truth table, find the Boolean function implemented by a 3-input minority gate. Simplify the function.	15
10.	Simplify $F = \Sigma (0, 1, 4, 5, 7, 9, 13)$ into sum-of-products form implement both of its form with necessary logic gates.	15
11.	Simplify $F = \Sigma (0, 1, 4, 5, 7, 9, 13)$ into product-of-sums form and implement both of its form with necessary logic gates.	15
12.	Let $F = A'C + A'B + AB'C + BC$ is a Boolean expression a) Express it in sum of minterms b) Find the minimal sum of products expression and implement both of its form with necessary logic gates.	15
13.	Viva-Voce	15

2025/01/14 11:36

Patuakhali Science & Technology University (PSTU)
Department of Computer Science and Information Technology (CSIT)

Final Examination: July-December 2020

Course Code: CCE 222 | Course Title: Digital Logic Design Sessional

Session: 2018-19, Program: B.Sc. Engg.(CSE), Semester: 4th

Marks – 70

[Answer the marked questions]

Simulation - 10/15

Section A (Use Simulator to Implement)

1. Design a combinational circuit which accepts a two-bit number and generates an output binary number equal to the cube of the input number. 20
2. Implement following Boolean function using exclusive or and AND gates
 $F = ABCD + A'B'C'D' + ABC'D' + A'B'CD$ 20
3. With 2×1 mux implement XOR gate and AND gate. 20
4. Design a D Flip-flop. 20
5. Design a T Flip-flop. 20
6. Design a circuit that compares two 4-bit numbers, A and B, to check if they are equal. The circuit has one output x, so that $x = 1$ if $A = B$, and $x = 0$ if $A \neq B$. 20

Section B (Use Circuit board to Implement by group work)

7. Design an adder that can add two decimal numbers as following. 30

$$\begin{array}{r} A B C D \\ + P Q R S \\ \hline W X Y Z \end{array}$$

8. Viva-Voce 20